

Q. 1) Greedy algorithm for a multi-Armed Bandit problem
Given that,

Machine A $\rightarrow [3, 2, 4]$

Machine B $\rightarrow [6, 7]$

Machine C $\rightarrow [5, 4, 6]$

formula:

$$q(a) = \frac{\text{Sum of rewards}}{N(a)}$$

Machine A:

$$q(A) = \frac{3+2+4}{3} = 3.0$$

Machine B:

$$q(B) = \frac{6+7}{2} = 6.5$$

Machine C:

$$q(C) = \frac{5+4+6}{3} = 5.0$$

The Greedy option is B = 6.5



Q.2) Greedy algorithm using incremental update method

Given,

Machine A [3, 2, 4]

Machine B [6, 7]

Machine C [5, 4, 6]

initially,

$$Q(A) = 0$$

$$N(A) = 0$$

$$Q(B) = 0$$

$$N(B) = 0$$

$$Q(C) = 0$$

$$N(C) = 0$$

Formula:

$$Q_{\text{new}}(a) = Q_{\text{old}}(a) + \frac{R - Q_{\text{old}}(a)}{N(a)}$$

Machine A: Rewards = 3, 2, 4

— first Reward = 3

initially:

$$Q(A) = 0$$

$$Q(B) = 0$$

$$N(A) = 1$$

$$\text{update } Q(A) = 0 + \frac{3 - 0}{1}$$

$$Q(A) = 3$$

Second R = 2

$$N(A) = 2$$

$$Q_{old}(A) = 3$$

$$Q(A) = 3 + \frac{2-3}{2}$$

$$Q(A) = 3 - \frac{1}{2}$$

$$Q(A) = 0$$

Third R = 4

$$N(A) = 3$$

$$Q_{old}(A) = 2.5$$

$$Q(A) = 2.5 + \frac{4-2.5}{3}$$

$$Q(A) = 2.5 + \frac{1.5}{3}$$

$$Q(A) = 3.0$$

$$Q_{new}(C) = 4.5$$

Third R = 6

$$N(C) = 3$$

$$Q_{old}(C) = 4.5$$

$$Q(C) = 4.5 + \frac{6-4.5}{3}$$

$$= 4.5 + \frac{1.5}{3}$$

final machine

$$N(A) = 3$$

$$Q(A) = 3.0$$

$$Q(C) = 5.0$$

final machine C

$$N(C) = 3$$

$$Q(C) = 5.0$$

Machine B (Rewards : 6, 7)

First R: 6

$$N(B) = 1$$

$$Q_{old}(B) = 0$$

Therefore

$$Q(B) = 0 + \frac{6 - 0}{1}$$

$$Q(B) = 6$$

$$Q(A) = 3.0$$

$$Q(B) = 6.5$$

$$Q(U) = 5.0$$

Second R: 7

$$N(B) = 2$$

$$Q(B) = 6 + \frac{7 - 6}{2}$$

$$Q_{old}(B) = 6$$

$$= 6 + \frac{1}{2}$$

$$Q(B) = 6.5$$

The highest value is

$$Q(B) = 6.5$$

final machine B

$$N(B) = 2$$

$$Q(B) = 6.5$$

Machine (Reward : 5, 4, 6)

first R: 5

$$N(C) = 1$$

$$Q_{old}(C) = 0$$

$$Q(New) = 5$$

second R: 4

$$N(C) = 2$$

$$Q_{old}(C) = 5$$